



Effects of laser-arc hybrid welding on microstructure and mechanical properties of dissimilar steel joint



Shoubing Yang ^a, Dongxu Wang ^a, Li Yang ^a, Fulong Zhang ^b, Chi Zhang ^c, Bin Zhou ^d, Chaojie Liu ^a, Genzhe Huang ^a

Show more

Outline | Share Cite

<https://doi.org/10.1016/j.ijleo.2022.169795>

[Get rights and content](#)

Full text access

Abstract

Laser-arc hybrid welding can improve production efficiency, and thus, it is being used in an increasing number of industries. Herein, the effects of laser-arc hybrid welding process parameters on the microstructure and mechanical properties of welded joint of low- and medium-carbon steels were investigated. Welding seam with best quality and no defects was obtained at welding current, arc voltage, laser power, and welding speed of 260A, 24.8V, 3300W, and 0.8m/min, respectively. Moreover, the enrichment of Mn in the weld zone promoted bainite formation. Rapid cooling led to the pseudo-eutectic phenomenon in the heat-affected zone (HAZ) on both sides of the weld; therefore, when an HAZ was closer to the weld on both sides, the pearlite content was higher. For low-carbon steel, the microstructure of the HAZ comprised ferrite, pearlite, and ferrite widmanstatten. For medium-carbon steel, the microstructure of the HAZ comprised a large amount of pearlite and a small amount of ferrite. In

welded joint of both types of steels, differences in microstructure and internal stress altered the nanostructure morphology and distribution characteristics of cementite. The welded joint also had considerably better tensile strength due to a large number of dislocations and better grain refinement, which led to tensile fractures in the base metal zone of low-carbon steel. For a prefabricated weld notch, the tensile fracture mode was a ductile fracture with massive dimples.



Keywords

Low- and medium-carbon steel; Laser-arc hybrid welding; Microstructure; Mechanical property

1. Introduction

Dissimilar steel welding can ensure effective use of the excellent properties of various materials, meet the application requirements of welded joint in complex environments, and reduce the production cost. Dissimilar steels are often welded using traditional metal inert/active gas welding [1], tungsten inert gas welding, and electron beam welding [2], [3]. However, traditional arc welding has some limitations, such as low welding efficiency, large heat input, grain coarsening in the heat-affected zone (HAZ), and severe welding deformation [4], [5], [6]. Electron beam welding is advantageous because of high energy density, small heat input, large penetration depth, and small deformation. However, it requires welding in a vacuum environment, which has limited its wide application[7].

Compared with traditional arc welding, laser welding is advantageous because it is characterized by small heat input, large penetration depth, small welding deformation, and high efficiency [8], [9]. In the past ten years, laser welding of dissimilar steels has developed rapidly. Dak Gaurav et al. [10] examined the effect of microstructure heterogeneity on the hardness and impact properties of welded joint of martensitic P92 steel/304L stainless steel obtained using laser welding. Chen et al. [11] considered 316L stainless steel/GH909 steel welded joint and examined the effects of different laser welding speeds on their microstructure and properties under low power and low welding speed, the temperature gradient of the weld pool was small,

which resulted in uniform distribution of weld elements and improved the uniformity of the microstructure of welded joint, resulting in improved tensile strength of welded joint. Gao et al. [12] studied the effect of laser heat input on the microstructure and mechanical properties of DP780/DP980 steel welded joint; laser welding significantly enhanced the microhardness of the weld zone (WZ). Zhang et al. [13] studied the effects of laser welding and laser oscillation welding on the microstructure and properties of PHS1500/PHS2000 steel welded joint. Compared with laser welding, laser oscillation welding resulted in excellent welded joint without welding defects, and it effectively inhibited the formation of δ -ferrite in the weld bead and led to the formation of a fully martensitic microstructure in the weld bead. Thus, while there have been advances in the laser welding of dissimilar steels, some disadvantages remain, such as weak gap bridging ability, complex workpiece assembly requirements, which limit the welding of thick plate dissimilar steel [14].

Laser-arc hybrid welding is an up-and-coming welding technology [8], and it combines the bridging connectivity of arc welding and the deep penetration of laser welding. It has the advantages of low heat input, large penetration depth, high welding efficiency [15], [16], [17]. Zhang et al. [18] studied the effect of element distribution on the formation of microstructure and mechanical properties of laser-arc hybrid welded joint of EH36 steel/316L stainless steel. Zhou et al. [19] studied the effect of laser and arc energy ratio on the microstructure and mechanical properties of 316L stainless steel/AH36 steel welded joint. A lower laser-to-arc energy ratio is helpful for reducing martensite content in the weld bead and increasing the number of large-angle grain boundaries. Spina et al. [20] studied the effect of laser-arc hybrid welding process parameters on the microstructure and mechanical properties of high-manganese TWIP steel (thickness=1.4mm)/dual-phase DP steel welded joint. The tensile strength of the welded joint was close to that of the DP steel base metal when heat input was in the range of 50–75kJ/m. This also illustrates that laser-arc hybrid welding has good applicability in thin plate welding. In summary, there have been some advances in studies on laser-arc hybrid welding of dissimilar steel. However, contemporary studies have focused on the relationship between the microstructure and mechanical properties of dissimilar steel welded joint, which have considered limited and simple methodologies.

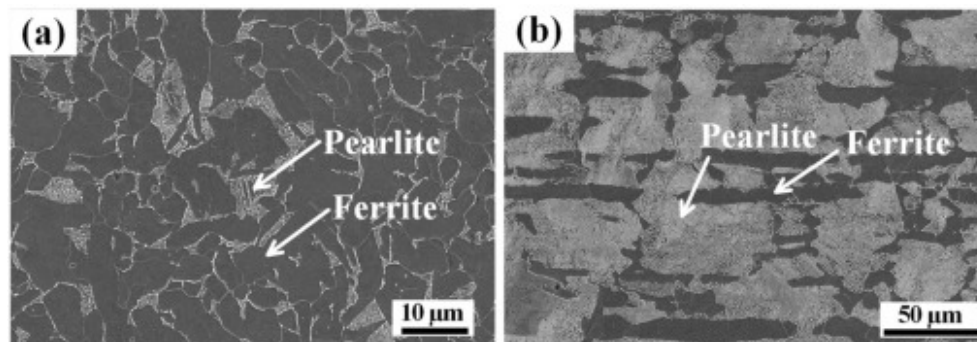
Few studies have investigated the relationship among the microstructure of dissimilar steel welded joint, grain growth orientation, nanostructures, and mechanical properties. Currently, low- and medium-carbon steel are widely used for manufacturing mechanical shafts and gears, constructing bridges, transportation, and other applications [21], [22]. In the manufacturing process, the welded joint of two types of steels should meet the production needs. In this study, for laser-arc hybrid welding of low- and medium-carbon steels, the effect of welding heat input on weld forming quality was explored.

Furthermore, the effects of microstructure formation mechanism, grain growth orientation, and nanostructure on the mechanical properties of welded joint were analyzed in detail.

2. Experimental materials and methods

2.1. Experimental materials

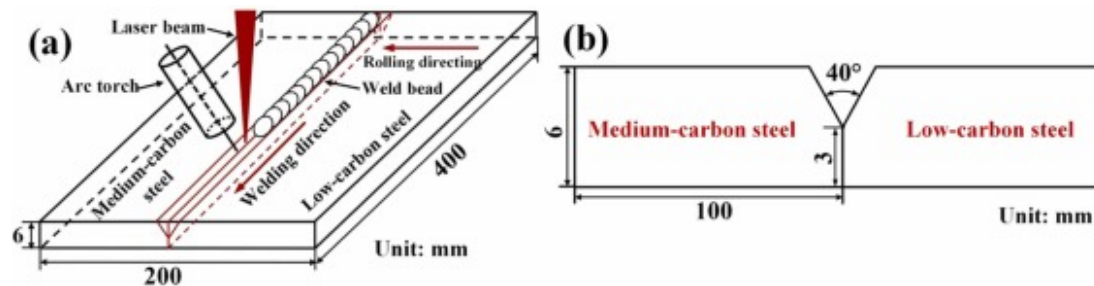
The test materials were rolled specimens of low- and medium-carbon steel. The microstructure of low-carbon steel comprises a large amount of granular ferrite and pearlite, and medium-carbon steel comprises ferrite and pearlite, as shown in Fig. 1. According to the testing scheme, the low- and medium-carbon steel specimens were cut into 400mm×100mm×6mm sized plates. The welding direction was perpendicular to the rolling direction, that is, along the long side of 400mm, as shown in Fig. 2(a). The groove was machined at a unilateral angle of 20°, the height of the blunt edge was 3mm, and the butt-joint gap between the steel plates was 0mm, as shown in Fig. 2(b). Lincoln RM-52 filler wire (diameter=1.2mm) was used. The chemical compositions and main mechanical properties of base metal and filler wire are shown in Table 1, Table 2, respectively.



[Download: Download high-res image \(297KB\)](#)

[Download: Download full-size image](#)

Fig. 1. Microstructure of base metals: (a) low-carbon steel; (b) medium-carbon steel.



[Download: Download high-res image \(135KB\)](#)

[Download: Download full-size image](#)

Fig. 2. Welding plate and welding sample groove dimensions: (a) welding plate size; (b) welding sample groove size.

Table 1. Chemical composition of base metals and filler wire (wt%).

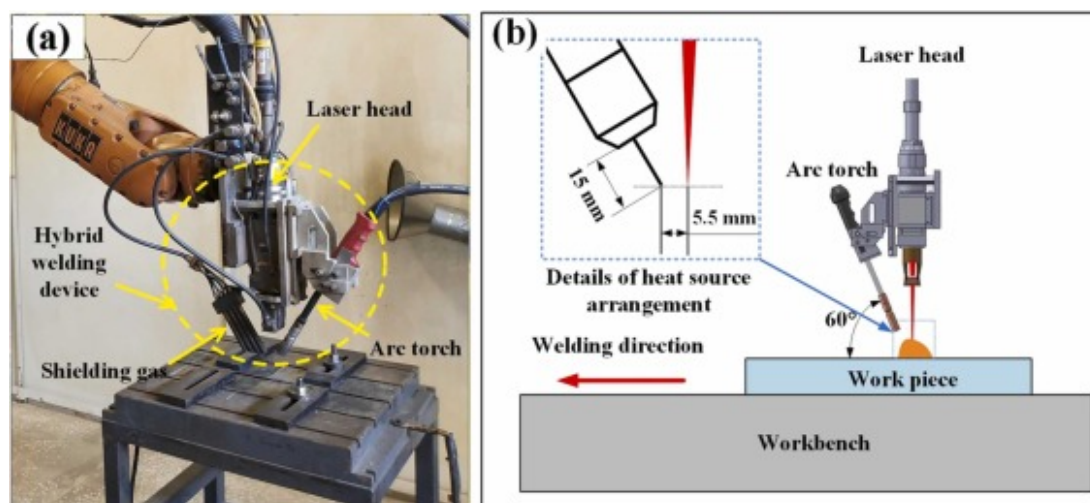
Materials	C	Si	Mn	P	S	Al	Cr	Ni	Cu	Fe
Low-carbon steel	0.19	0.18	0.37	0.02	0.01	0.03	0.02	0.02	—	Bal
Medium-carbon steel	0.49	0.22	0.59	0.02	—	0.01	0.02	0.01	0.01	Bal
RM-52	0.06	0.88	1.49	0.02	0.01	—	—	—	0.14	Bal

Table 2. Main mechanical properties of base metal and filler wire.

Materials	Yield strength (MPa)	Tensile strength (MPa)	Elongation (%)
Low-carbon steel	275	497	28.0
Medium-carbon steel	400	663	24.5
RM-52	465	560	28.0

2.2. Experimental methods

A HL4006D Nd:YAG laser (TRUMPF, Germany) was used in this study. The maximum output power was 4 kW, laser wavelength was 1.06 μm , and spot diameter was 0.6 mm. For arc welding, a YM-350AG2 semi-automatic MAG welding machine was used (Panasonic, Japan). The laser-arc hybrid welding device was driven by a six-axis robot (KUKA) to implement automatic welding. The hybrid welding experimental device is shown in Fig. 3(a). The welding heat source was arranged with the arc in front and the laser beam at the back. The inclination angle between the welding torch and the horizontal plane was 60° , and the angle between the welding torch and the laser beam was 30° . The defocus of the laser beam was -2 mm below the surface of the welded plates, the distance between the laser beam and the end of the filler wire at the surface of the welded plates was 5.5 mm, and the extension length of the filler wire was 15 mm. Fig. 3(b) shows a schematic representation of the welding process. The shielding gas used during welding was a mixture of 90% Ar and 10% CO_2 , and the output flow was 18 L/min. Fixed welding parameters (presented in Table 3) were used. The different values of welding current, arc voltage, laser power, and welding heat input used in the experiments are presented in Table 4, Table 5. Before welding, oxide scales on the surfaces of the welding plates were polished and removed using a grinder. An acetone solution was used to remove oil stains on the surfaces of the welding specimens.



[Download: Download high-res image \(243KB\)](#)

[Download: Download full-size image](#)

Fig. 3. Laser-arc hybrid welding device and welding schematic diagram: (a) laser-arc hybrid welding device; (b) welding schematic diagram.

Table 3. Laser-arc hybrid welding parameters.

Welding parameters	Values
Welding speed, V [m/min]	0.8
Defocus amount, Δf [mm]	-2
Distance between laser and arc, D_{LA} [mm]	5.5
Angle of welding torch, θ [°]	60
Wire diameter, d [mm]	1.2
Wire extension length, L [mm]	15
Shielding gas flow rate, γ [L/min]	18

Table 4. Varying welding current, arc voltage, and total welding heat input.

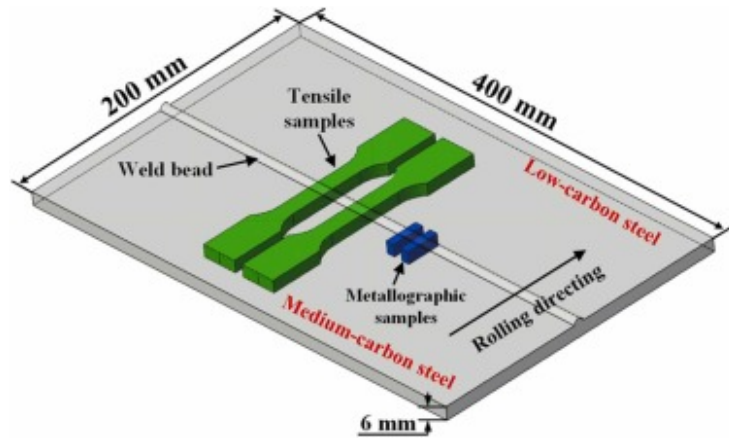
Welding current, I [A]	Arc voltage, U [V]	Arc heat input, Q_1 [kJ/m]	Laser powder, P [W]	Laser heat input, Q_2 [kJ/m]	Total Heat input, Q [kJ/m]
200	21.6	324.0	3100	232.5	556.5
220	22.8	376.2	3100	232.5	608.7
240	23.6	424.8	3100	232.5	657.3
260	24.8	483.6	3100	232.5	716.1
270	26.4	534.6	3100	232.5	767.1

Table 5. Varying laser power and total welding heat input.

Laser powder, P [W]	Laser heat input,		Arc voltage, U [V]	Arc heat input,	
	Q_2 [kJ/m]	Welding current, I [A]		Q_1 [kJ/m]	Total Heat input, Q [kJ/m]
2900	217.5	260	24.8	483.6	701.1
3100	232.5	260	24.8	483.6	716.1
3300	247.5	260	24.8	483.6	731.1
3500	262.5	260	24.8	483.6	746.1
3700	277.5	260	24.8	483.6	761.1

After the welding was complete, from each set of parameter welding samples, two sets of metallographic samples and tensile samples were cut using a wire cutting machine. The laser beam did not form a stable laser keyhole in the molten pool in the initial welding arc stage, and the welding current and arc voltage was not sufficiently steady in the arc extinguishing stage; therefore, the metallographic and tensile samples were obtained from the middle of the welding plates, as shown in Fig. 4. Metallographic samples were inlaid with an inlaying machine, polished with sandpaper and a diamond polishing agent, cleaned with absolute ethanol, and then, corroded with a 4% nitric alcohol solution after drying. Optical microscopy (OM; Leica DM2700M) was used to observe and analyze the cross-section morphologies of the welds. Scanning electron microscopy (SEM; JSM-6510 F) and self-contained energy dispersive spectrometry (EDS) were used to observe and analyze the microstructure and elemental distribution of the welded joint, respectively. Differences in the mechanical properties of different areas of welded joint were examined using electron backscatter diffraction (EBSD; the Nordlys Max3), and it was also used to measure the grain sizes and analyze the orientation difference between grains at different positions of the welded joint. The analysis positions are presented in Fig. 5. To further investigate the root cause of the differences in the mechanical properties of different regions of the welded joint from a nanoscopic perspective, transmission electron microscopy (TEM; Tecnai G2 F20) was used to analyze the microstructure and dislocations in different areas of the welded joint (analytical positions shown in Fig. 5). A Vickers microhardness tester (MH-06) was used to measure the hardness of different areas of the welded joint. The hardness test load was 500 gf, the loading time was 10 s, the interval of hardness indentation was 0.25 mm, and the hardness measurement position

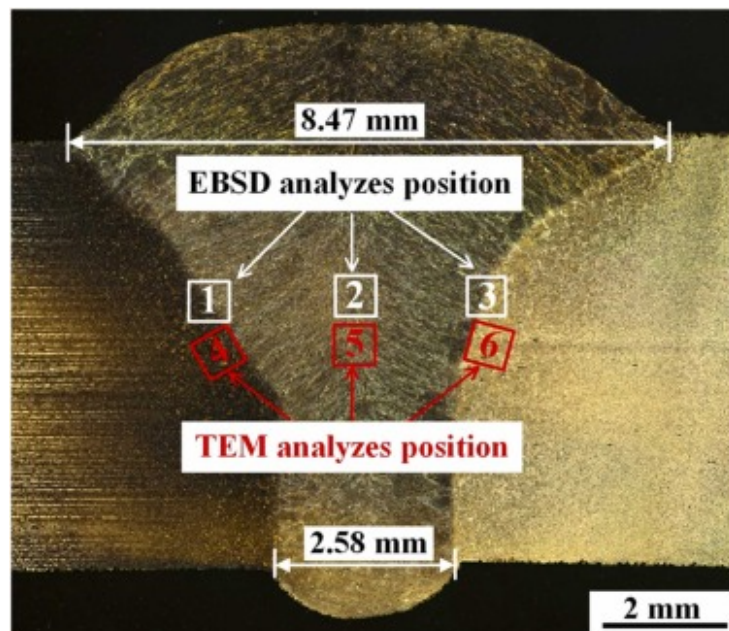
was 3 mm from the upper edge of the weld. Tensile properties of the welded joint were examined using an electronic tensile testing machine. The sizes of the tensile test specimens are shown in Fig. 6(a). To examine the tensile strength of the weld bead, a semi-circular notch (radius=3 mm) was made on the weld bead to ensure that the tensile sample broke at the weld bead (Fig. 6(b)).



[Download: Download high-res image \(111KB\)](#)

[Download: Download full-size image](#)

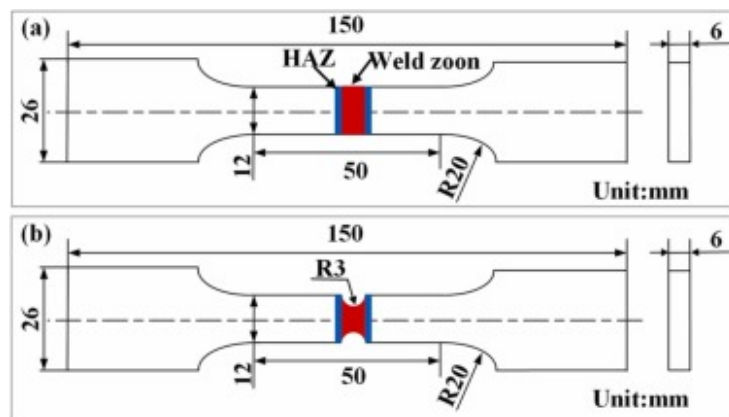
Fig. 4. Extraction scheme of metallographic and tensile specimens.



[Download: Download high-res image \(367KB\)](#)

[Download: Download full-size image](#)

Fig. 5. EBSD and TEM analysis positions of welded joint for with optimal welding parameters (welding current 260 A, arc voltage 24.8 V, laser power 3300 W, and welding speed 0.8 m/min).



[Download: Download high-res image \(100KB\)](#)

[Download: Download full-size image](#)

Fig. 6. Sizes of tensile specimens: (a) normal tensile specimen; (b) prefabricated weld notched tensile specimen.

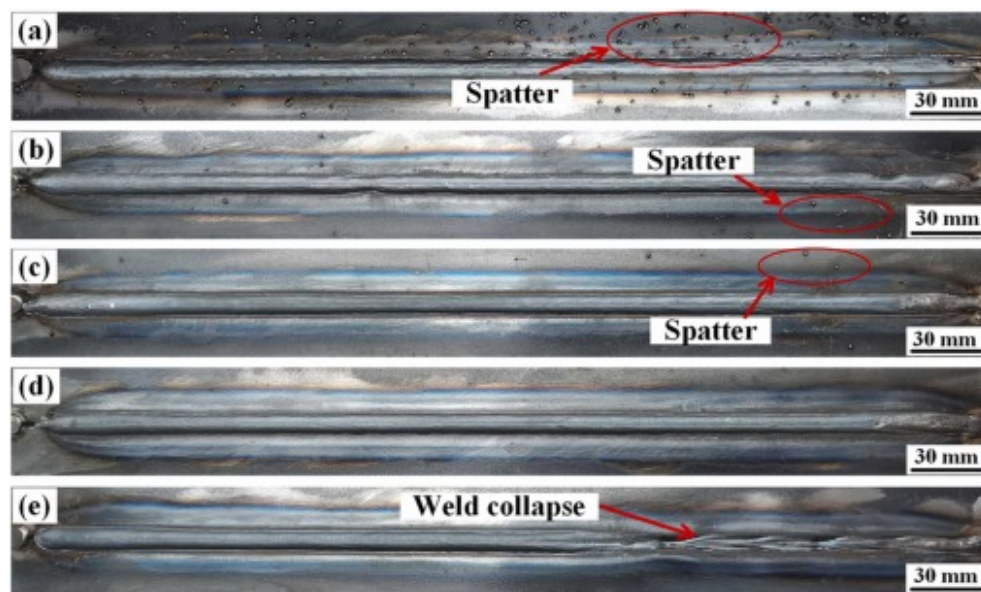
3. Results and discussion

3.1. Effects of welding heat input on weld forming quality

3.1.1. Effects of arc heat input on weld forming quality

Fig. 7 shows the effects of different arc heat inputs on the weld surface morphology. When the arc heat input was 483.6 kJ/m, the weld surface is smooth. When the arc heat input was less than 483.6 kJ/m, the continuity of the weld surface worsened, and a large amount of spatter was generated around the weld. This phenomenon occurred because the heat input of the arc was relatively low, which hindered complete combustion of the arc. Because of this, the droplet could not break away from the front end of the filler wire under the given welding current heat input. The droplet was in the form of an unstable short-circuit transition, which generated spatters. This led to a poorly formed weld surface. With increase in arc heat input, the electromagnetic force and plasma flow force on the droplet increased, which promoted the transition of the droplet from the tip of the welding wire to the molten pool. Therefore, a stable droplet transfer mode was generated, thus improving the quality of weld formation. When the arc heat input reached 534.6 kJ/m (as shown in Fig. 7(e)), excessive penetration was observed on the

weld. This was attributed to excessively high arc heat input and considerable melting in the base metal, which resulted in the collapse of the molten pool during welding. Thus, the arc heat input should be reasonably selected to improve the quality of the formed weld surface.

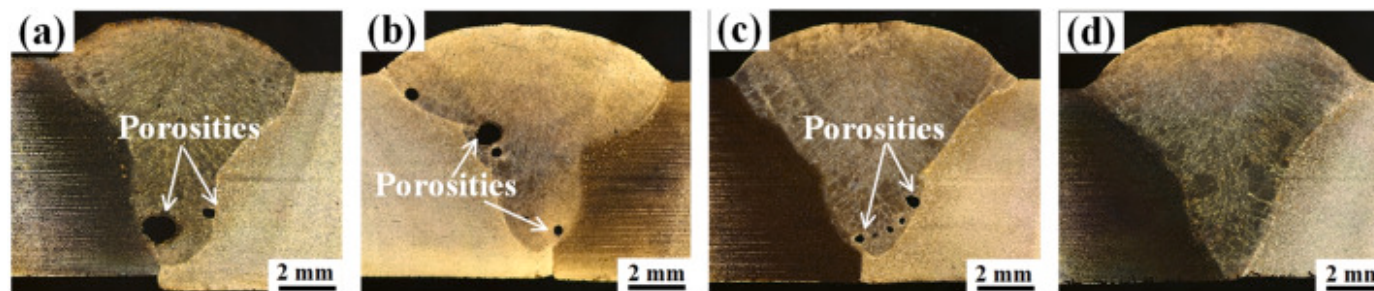


[Download: Download high-res image \(420KB\)](#)

[Download: Download full-size image](#)

Fig. 7. Effects of different arc heat inputs (Q_1 , kJ/m) on weld surface morphology at laser heat input of 232.5 kJ/m: (a) 324.0; (b) 376.2; (c) 424.8; (d) 483.6; (e) 534.6.

Fig. 8 shows the effect of different arc heat values on the cross-section morphology of the weld beads. When the arc heat was 534.6 kJ/m, the welding seam appears sunken as the molten pool liquid was flowed out from the back of the welding plate, so it was not considered for further analysis. Fig. 8(a)–(c) show that when the arc heat input was 324, 376.2, and 424.8 kJ/m respectively, the weld bead was not completely penetrated. With increase in arc heat input, the cross-sectional morphology of the weld changed from goblet-shaped to round nail-shaped. This is because the penetration ability of the arc energy was limited, because of which, the welding heat was concentrated in the upper half of the weld bead. Thus, the weld bead could not be completely penetrated when low arc heat input was used. At a constant laser heat input, increasing the arc heat input increased the amount of melted metal in the upper half of the weld, which changed the weld cross-section shape from goblet to round nail.



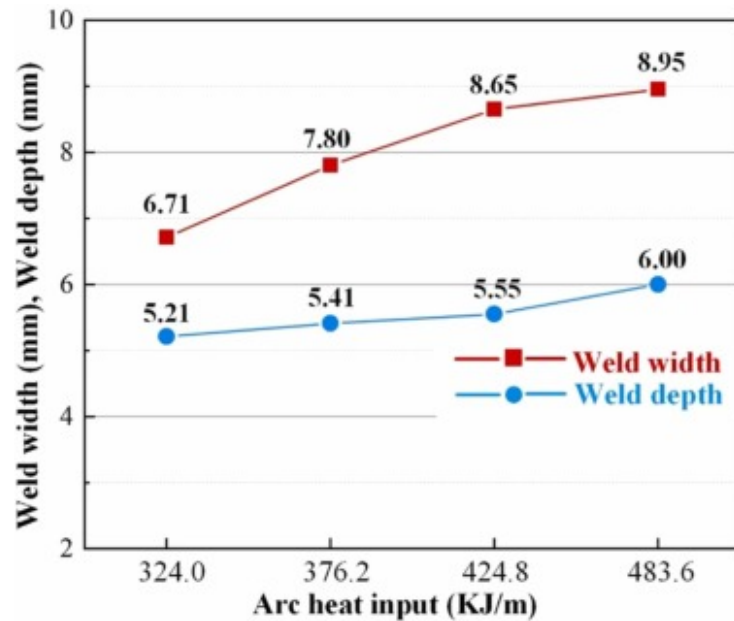
[Download: Download high-res image \(699KB\)](#)

[Download: Download full-size image](#)

Fig. 8. Effects of different arc heat inputs (Q_1 , kJ/m) on weld cross-section morphology at laser heat input of 232.5 kJ/m: (a) 324.0; (b) 376.2; (c) 424.8; (d) 483.6.

The cross-sectional morphology of the weld (Fig. 8) shows that at arc heat input of 324, 376.2, and 424.8 kJ/m, welding porosity defects appeared in the cross-sectional morphology of the welds. At arc heat input of 483.6 kJ/m, there was no porosity in the weld section, as shown in Fig. 8(d). This is because when the arc heat input was less than 483.6 kJ/m, the total heat input of the weld was small, and the cooling rate of the weld was fast. When the upward speed of the bubbles generated at the bottom of the molten pool was less than the solidification speed of the molten pool, they were trapped in the weld, forming welding porosities. At arc heat input of 483.6 kJ/m, the welding seam was completely penetrated. The bubbles in the molten pool floated to the top of the molten pool and also escaped from the bottom of the molten pool because of the flow of liquid metal inside the molten pool, which considerably increased the probability of the bubbles escaping the pool. Therefore, at arc heat input of 483.6 kJ/m, there were no porosity defects in the weld section.

Cross-sectional dimensions of the welds (weld width and depth) were measured at arc heat inputs of 324, 376.2, 424.8, and 483.6 kJ/m (Fig. 9). With increase in arc heat input, weld width and depth increased gradually. As the arc heat input increased from 324 to 483.6 kJ/m, the weld width and weld depth increased by 2.24 mm and 0.79 mm, respectively. The variations in weld width and depth show that the influence of arc heat input was more significant on weld width. This is because during laser-arc hybrid welding, the heat input of the arc is mainly concentrated on the upper half of the weld, which increases the melting of the base metal on the weld surface, thus increasing the weld width.



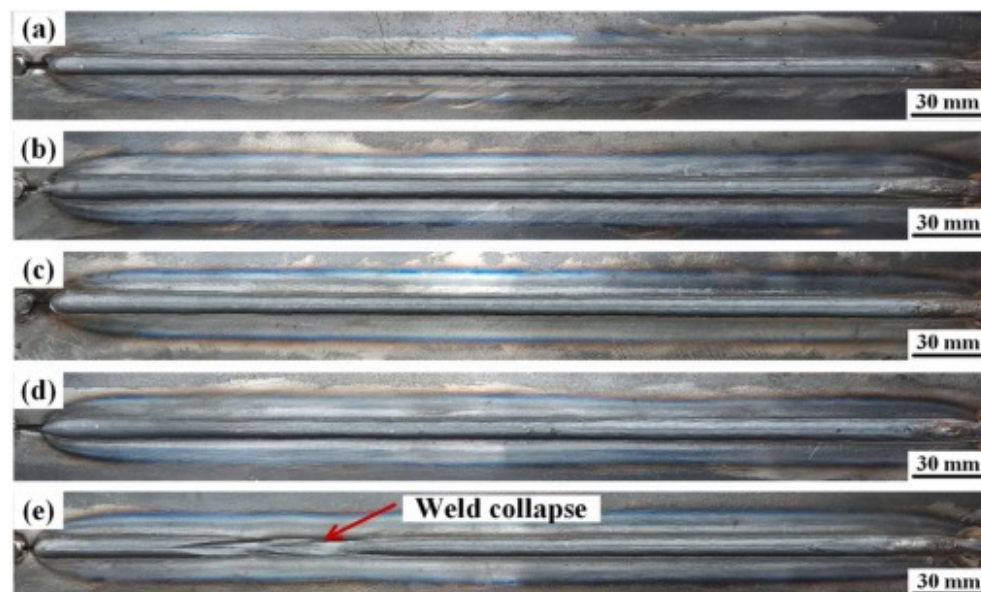
[Download: Download high-res image \(108KB\)](#)

[Download: Download full-size image](#)

Fig. 9. Effect of different arc heat input on weld width and depth at laser heat input of 232.5 kJ/m.

3.1.2. Effects of laser heat input on weld forming quality

The effects of different arc heat input on the weld seam forming quality show that when arc heat input was 483.6 kJ/m, the welding seam surface was smooth, the welding seam reached full penetration, and there were no porosity defects. Therefore, the influence of laser heat input on the welding seam forming quality was investigated at arc heat input of 483.6 kJ/m. Fig. 10 shows the effects of different laser heat values on the surface morphologies of the weld. The different values of laser heat only slightly influenced the surface morphologies of the weld bead, and the weld formation was good. At laser heat input of 277.5 kJ/m, the weld bead penetrated welding plates and collapsed (Fig. 10 (e)). This is because the energy output of the laser is too high, which causes penetration of the base metal and the molten liquid metal flowed to the back of the plates.

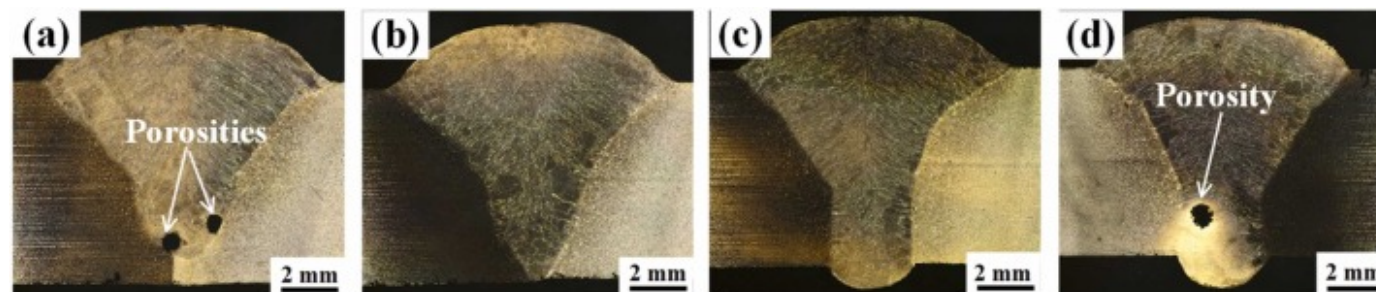


[Download: Download high-res image \(400KB\)](#)

[Download: Download full-size image](#)

Fig. 10. Effects of different laser heat input (Q_2 , kJ/m) on the surface morphologies of the weld at arc heat input of 483.6 kJ/m: (a) 217.5; (b) 232.5; (c) 247.5; (d) 262.5; (e) 277.5.

Fig. 11 shows the cross-sectional morphologies of the welds under different laser heat inputs. At laser heat input of 217.5 kJ/m, the weld bead was not completely penetrated. When laser heat input was more than 232.5 kJ/m, all weld beads were completely penetrated. With increase in laser heat input, the shape of the cross-section of the weld gradually changed from round nail to goblet shape. At laser heat input of 247.5 kJ/m, the cross-sectional morphology of the weld bead was the most similar to that obtained using laser-arc hybrid welding. At this time, the arc area of the weld was clearly distinguished from the laser area, and there was excellent weld formation. With increase in laser heat input, the synergistic effect of the laser and the arc was enhanced, and the laser keyhole effect became more prominent. Therefore, the laser beam generated deep penetration welding at the bottom of the weld bead. With increase in laser heat input, the cross-sectional morphology of the weld bead gradually changed to a goblet shape. Moreover, the cross-sectional morphologies of the weld (Fig. 11) show that porosities appeared in the weld section at laser heat input of 217.5 and 262.5 kJ/m. The causes for porosity formation are similar to the causes for the occurrence of porosity in the weld section under different arc heat values in Section 3.1.1.

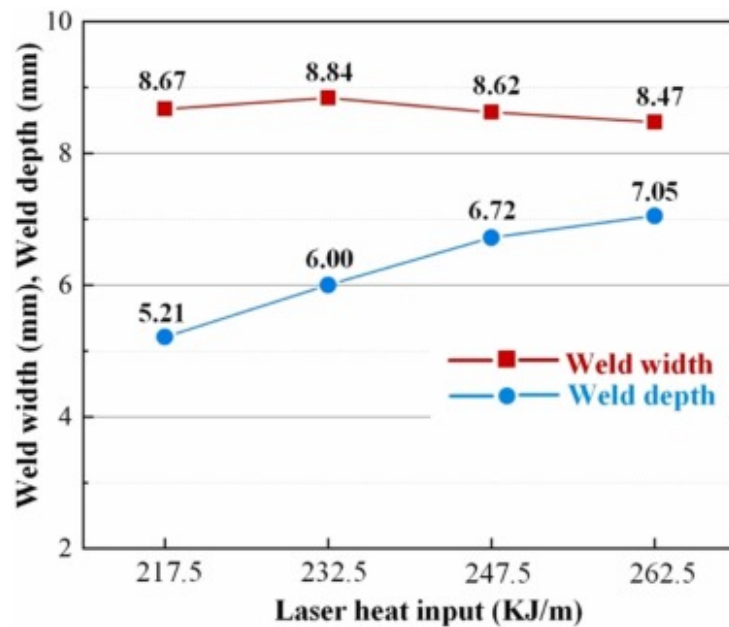


Download: [Download high-res image \(293KB\)](#)

Download: [Download full-size image](#)

Fig. 11. Effect of different laser heat values (Q_2 , kJ/m) on the cross-section morphology of the weld at arc heat input of 483.6 kJ/m: (a) 217.5; (b) 232.5; (c) 247.5; (d) 262.5.

To examine the effect of different laser heat values on the cross-sectional size of the weld bead, the width and depth of the weld under different laser heat inputs were measured (Fig. 12). With increase in laser heat input, the weld width first increased and then decreased. This is because the weld was not completely penetrated when the laser heat input was less than 232.5 kJ/m. The molten liquid metal of the base metal and the welding wire filled the surface of the weld bead, increasing the width of the weld. At laser heat input of 247.5 kJ/m, the weld bead was completely penetrated. At this time, part of the molten metal flowed into the back of the weld bead, forming the backside of the weld bead, which decreased the amount of metal filled on the upper surface of the weld; therefore, the width of the weld bead was decreased. The weld depth curve shows that with increase in laser heat input, the weld depth gradually increased. When laser heat input was increased from 217.5 to 262.5 kJ/m, the maximum increments of weld bead width and depth were 0.37 mm and 1.84 mm, respectively. The variations in the width and depth of the weld bead show that the laser heat input had a more significant effect on the weld bead depth. This is because the increase in laser heat input enhanced the effect of laser on arc stabilization, the effect of laser keyhole became more prominent, and the weld penetration increased considerably.



[Download: Download high-res image \(106KB\)](#)

[Download: Download full-size image](#)

Fig. 12. Effect of different laser heat inputs on weld width and depth at arc heat input of 483.6 kJ/m.

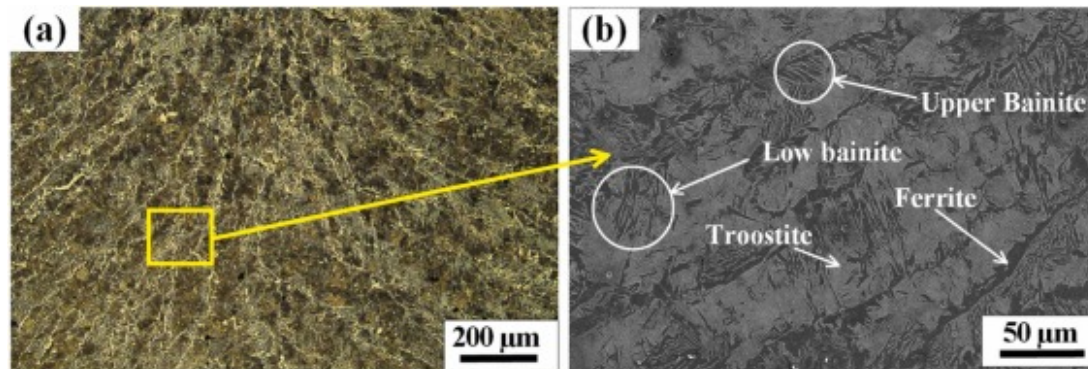
From the influence of different arc and laser heat inputs on the welding seam forming quality, it can be concluded that at arc and laser heat input of 483.6 kJ/m and 247.5 kJ/m, the best welding seam forming quality was obtained, and it showed the typical morphology characteristics of laser-arc hybrid welding. Therefore, the optimal welding parameters, corresponding to the heat input, are $I = 260$ A, $U = 24.8$ V, $P = 3300$ W, and $V = 0.8$ m/min.

3.2. Microstructural analyses of welded joint

The welding joint of low- and medium-carbon steel obtained using laser-arc hybrid welding include the WZ, low-carbon steel HAZ, and medium-carbon steel HAZ. In this section, the welded joint obtained using the optimal welding parameters ($I = 260$ A, $U = 24.8$ V, $P = 3300$ W, $V = 0.8$ m/min) was considered for further investigating its microstructure formation mechanism and mechanical properties.

3.2.1. Microstructure of WZ

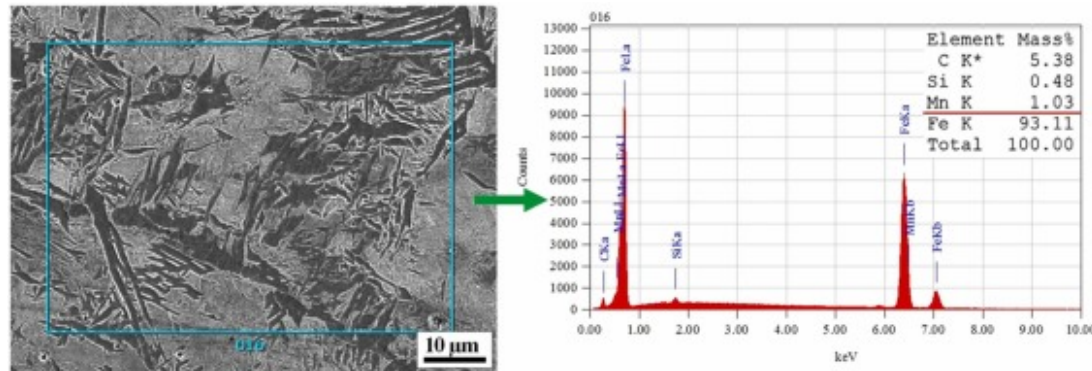
Fig. 13 shows the microstructure image of the WZ. The enlarged SEM image of WZ in Fig. 13(b) shows that the microstructure of the WZ comprises acicular ferrite, troostite, upper bainite, and lower bainite. The WZ has a large amount of bainite due to the large amount of Mn (1.49 wt%) in the filler wire. According to the EDS element analysis at the weld center (as shown in Fig. 14), there was high Mn content in the WZ (1.03 wt%). Compared with Mn content in the base metal (0.370–0.587 wt%), Mn content in the WZ was considerably higher, which improved the stability of austenite, and decreased the transformation temperature of austenite to that of ferrite. Moreover, Mn has a certain effect on reducing the activity of carbon atoms in steel [23], [24]; it inhibits the diffusivity of carbon atoms, shifts the continuous cooling temperature curve to the right, and decreases the critical cooling rate formed by the transformation of bainite. Therefore, a large amount of bainite structure was generated during the rapid cooling of the WZ.



Download: [Download high-res image \(463KB\)](#)

Download: [Download full-size image](#)

Fig. 13. Microstructure of WZ under (a) OM and (b) SEM.



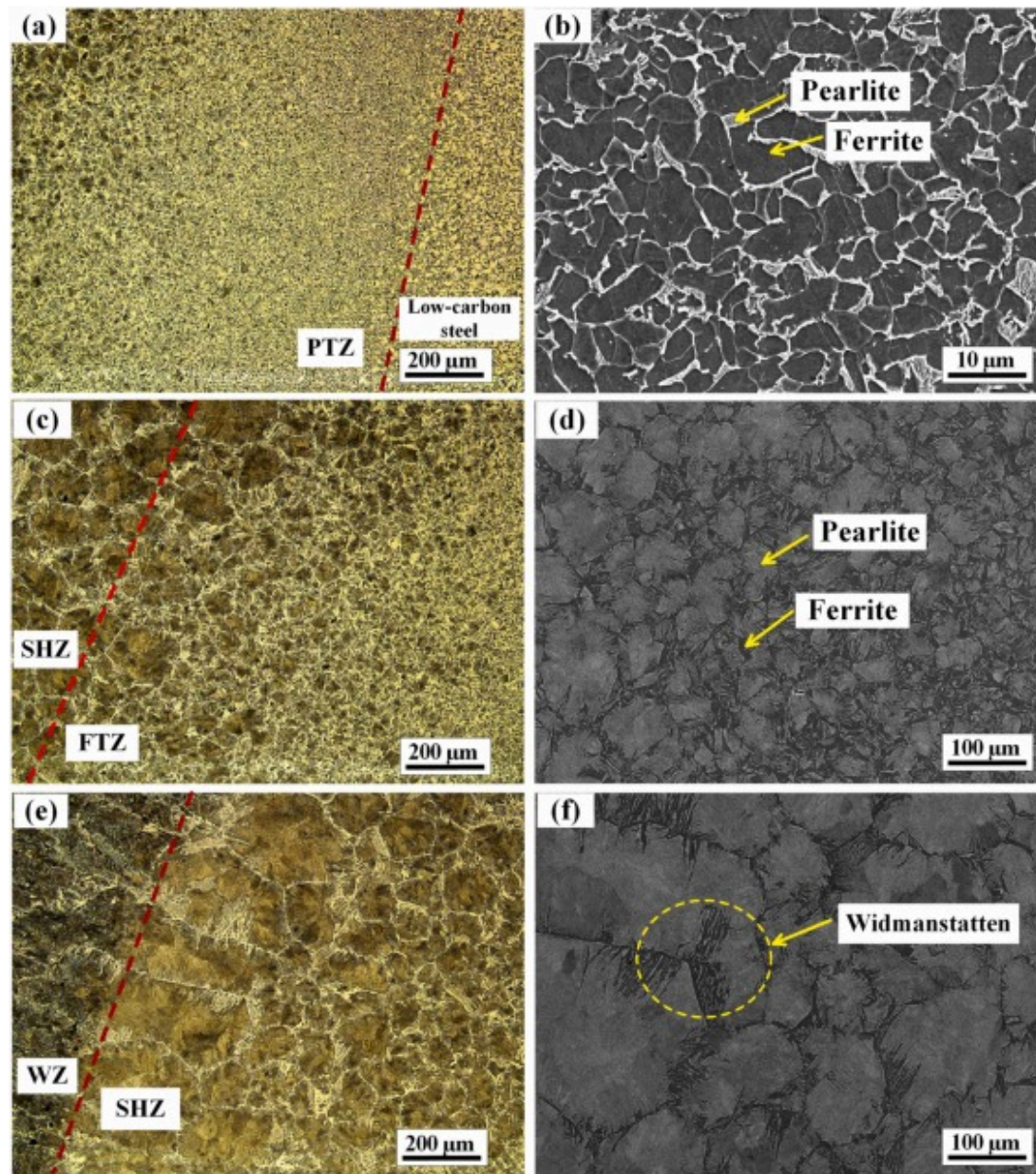
[Download: Download high-res image \(283KB\)](#)

[Download: Download full-size image](#)

Fig. 14. EDS elemental analysis of the WZ.

3.2.2. Microstructure of low-carbon steel HAZ

According to the different heating temperatures, the HAZ of low-carbon steel can be divided into partly transformation zone (PTZ), fully transformation zone (FTZ), and superheating zone (SHZ) (Fig. 15). PTZ was heated between A_{c1} and A_{c3} during the welding heating process (Fig. 15 (a)). FTZ was heated between A_{c3} and $1000\text{ }^{\circ}\text{C}$, and the microstructure was transformed entirely at this zone (Fig. 15 (c)). The temperature of the HAZ near the fusion line exceeded $1100\text{ }^{\circ}\text{C}$ and below the solidus line. This part of the HAZ is called the SHZ (Fig. 15 (e)).



[Download: Download high-res image \(2MB\)](#)

[Download: Download full-size image](#)

Fig. 15. Microstructure images of low-carbon steel HAZ: (a, b) PTZ; (c, d) FTZ; (e, f) SHZ.

The SEM microstructure image of the PTZ of low-carbon steel is shown in Fig. 15 (b). The microstructure of the PTZ comprises ferrite and pearlite. The grains in the PTZ are more refined than those in the base metal (i.e., low-carbon steel) (Fig. 15 (a)). In the PTZ range, pearlite in low-carbon steel is fully austenitized, while ferrite is retained because the temperature did not reach the austenite transformation temperature. In the subsequent rapid cooling process, austenite is transformed into fine ferrite and pearlite, thus producing refined grains in this region.

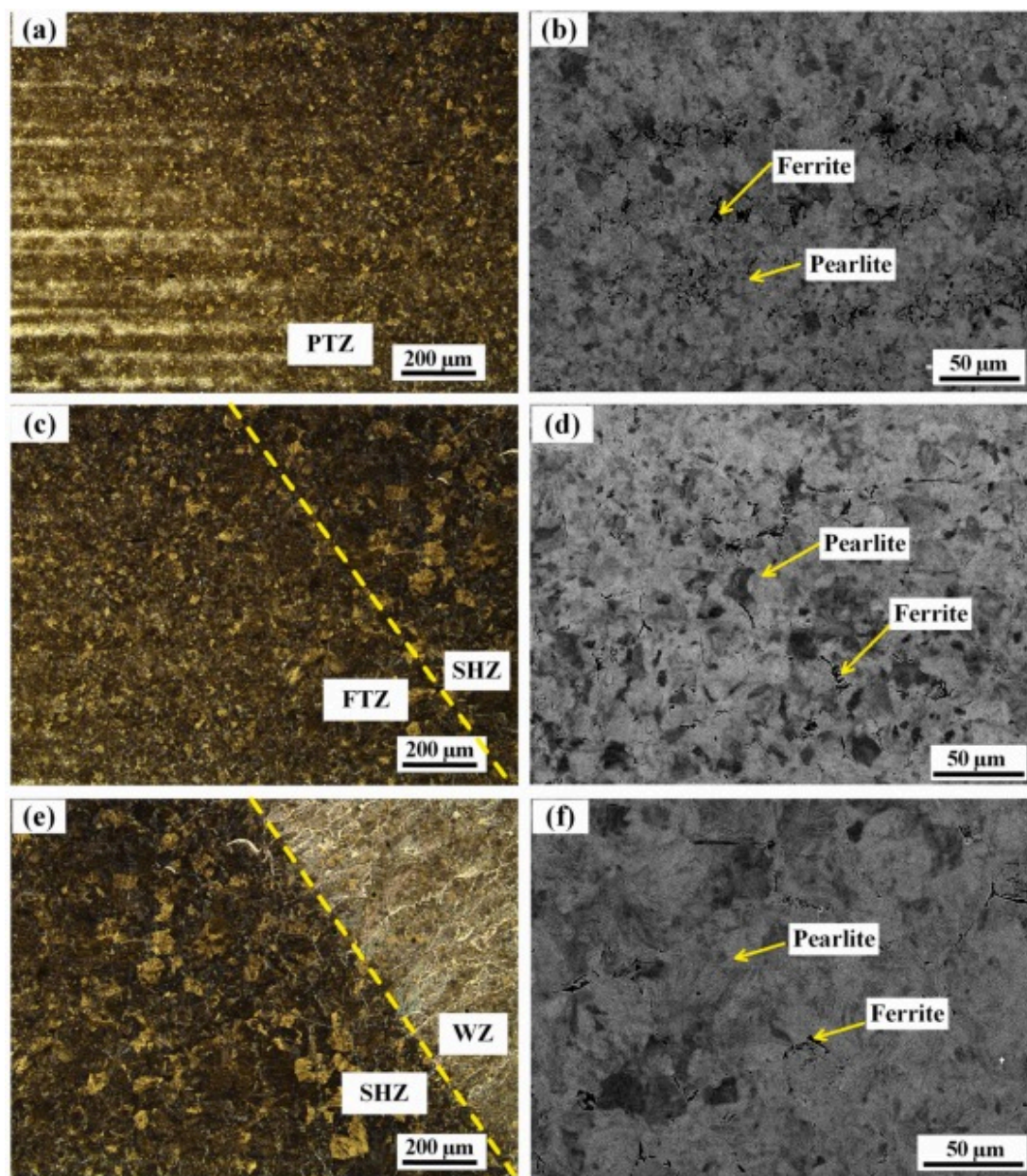
The SEM microstructure image of the FTZ of low-carbon steel is shown in Fig. 15 (d). The microstructure of the FTZ comprises ferrite and pearlite. The microstructure image obtained using low-power OM in FTZ (Fig. 13 (c)) shows that closer to the weld bead area, the pearlite content is higher and the pearlite grains are coarser. This is because faster cooling rate of the HAZ closer to the WZ results in larger shifting of the eutectic point, and it leads to the pseudo-eutectic phenomenon.

Fig. 15 (f) shows the SEM microstructure image of the SHZ of low-carbon steel. The microstructure of the SHZ comprises the ferrite widmanstatten structure; coarser grains are observed near the WZ (Fig. 15 (e)). This is because the SHZ is close to the WZ. When the temperature increases above the ferrite-austenite transformation line, the ferrite and pearlite in the low-carbon steel structure are entirely transformed into fine austenite structures. With further increase in temperature, the austenite is superheated to form a coarse austenite structure. In the subsequent rapid cooling process, the ferrite grows along a specific crystal plane of coarse austenite (habitual plane $\{111\text{ A}\}$) and in a certain crystal degree direction to form acicular proeutectoid ferrite. Part of this acicular proeutectoid ferrite precipitated in a network around the coarse austenite grain boundary, and some part precipitated independently in acicular shape from the austenite grain boundary into the grain by a shear mechanism. Carbon atoms of ferrite connected with the austenite grain boundaries, diffusing into the parent austenite on both sides; thus, the concentration of austenite carbon between the acicular ferrite increased continuously. Finally, the pearlitic microstructure was formed during the cooling process. The widmanstatten structure had coarse grains as well as a large number of ferrite needles, forming fragile surfaces, which increased the hardness and decreased the toughness of the SHZ.

3.2.3. Microstructure of medium-carbon steel HAZ

Similar to low-carbon steel HAZ, medium-carbon steel HAZ was divided into PTZ, FTZ, and SHZ (Fig. 16 (a), (c), and (e)). The microstructure of the PTZ of medium-carbon steel comprises fine ferrite and pearlite and refined grains (Fig. 16 (b)). Both the FTZ and SHZ comprise pearlite and ferrite (Fig. 16 (d) and (f)). Moreover, the pearlite content increases in the region closer to the WZ. In the SHZ, it almost entirely comprises pearlite, and there is only a small amount of ferrite. This is attributed to the faster cooling

rate of the HAZ near the WZ, which caused various shifts in the eutectic point, resulting in a pseudo-eutectic structure. Therefore, almost all austenite was transformed into pearlite at high temperatures, and only a small amount was transformed into ferrite.



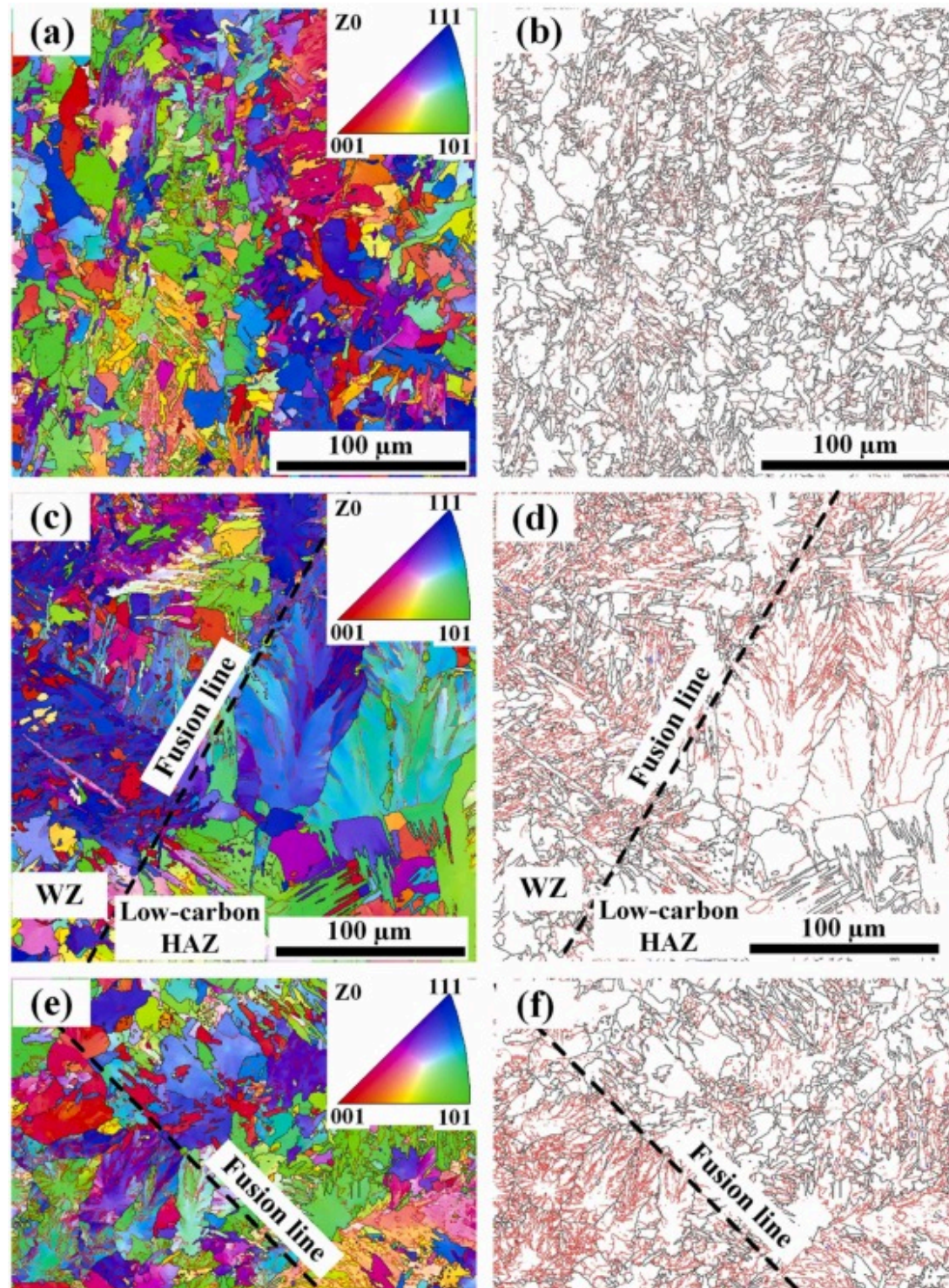
[Download: Download high-res image \(2MB\)](#)

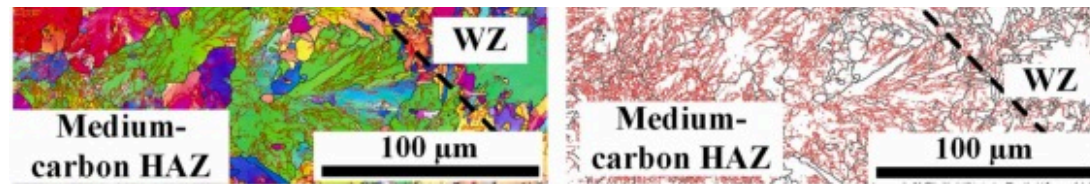
[Download: Download full-size image](#)

Fig. 16. Microstructure images of the HAZ of medium-carbon steel: (a, b) PTZ; (c, d) FTZ; (e, f) SHZ.

3.3. EBSD analysis of welded joint

To examine the grain size and grain boundary orientation in different regions of the welded joint, the welded joint was examined using EBSD. Fig. 17 shows EBSD map and grain boundary map of the WZ, near the fusion line on the low-carbon steel side, and near the fusion line on the medium-carbon steel side. The color difference inside the grain and the grain boundary diagram in EBSD images of the three regions shows that there are many fine subgrains with same and different orientations inside the grain. Fig. 17 (d) and (f) show that the number of subgrains in the HAZ on the low-carbon steel side is significantly lower than that in the WZ, and the number of subgrains in the HAZ on the medium-carbon steel side is significantly higher than that in the WZ. The existence of a large number of subgrains in the three regions helps strengthen the welded joint. The grain boundary angles of the WZ, near the fusion line on one side of low-carbon steel, and near the fusion line on one side of medium-carbon steel were counted (Fig. 18). The grain boundary was defined as a large-angle grain boundary when the grain boundary orientation difference was greater than 15° . The number of grains with large-angle grain boundary in the WZ accounted for approximately 62%, near the fusion line on one side of low-carbon steel accounted for approximately 47%, and near the fusion line on one side of medium-carbon steel accounted for approximately 36%. From the perspective of data points, there are a large number of large-angle grain boundaries in the WZ, indicating that most of the grains in the WZ are adjacent to the large-angle grain boundaries. For polycrystalline materials, higher number of large-angle grain boundaries in the WZ resulted in greater degree of mismatch between grains, which is more conducive to preventing and pinning the dislocations between grains or forming a “shielding wall” of dislocations at the grain boundaries. Therefore, the large-angle distribution of grains in WZ plays an important role in strengthening the WZ.

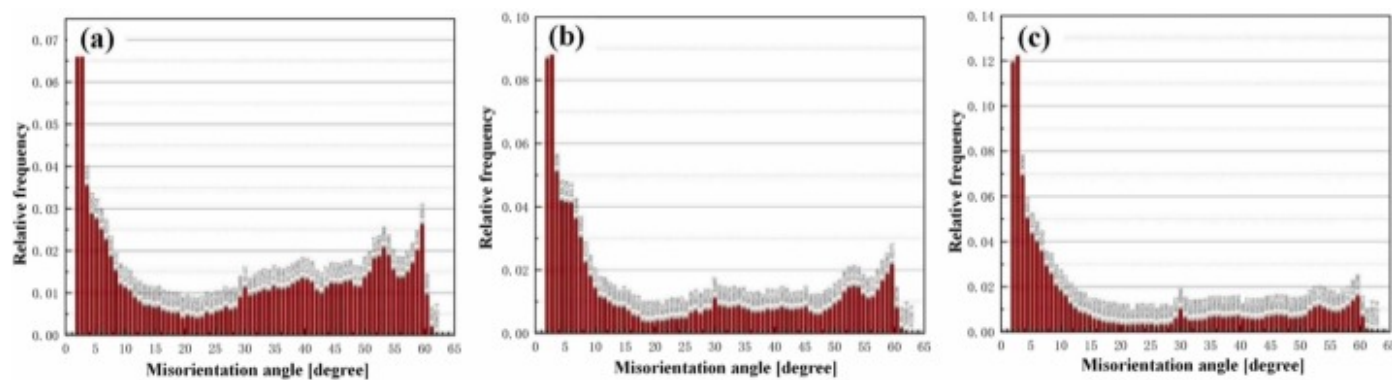




[Download: Download high-res image \(2MB\)](#)

[Download: Download full-size image](#)

Fig. 17. EBSD map and grain boundary map of different regions of the welded joint: (a, b) WZ; (c, d) near the fusion line of low-carbon steel; (e, f) near the fusion line of medium-carbon steel.



[Download: Download high-res image \(193KB\)](#)

[Download: Download full-size image](#)

Fig. 18. Grain boundary angles in different regions of the welded joint: (a) WZ; (b) near the fusion line of low-carbon steel; (c) near the fusion line of medium-carbon steel.

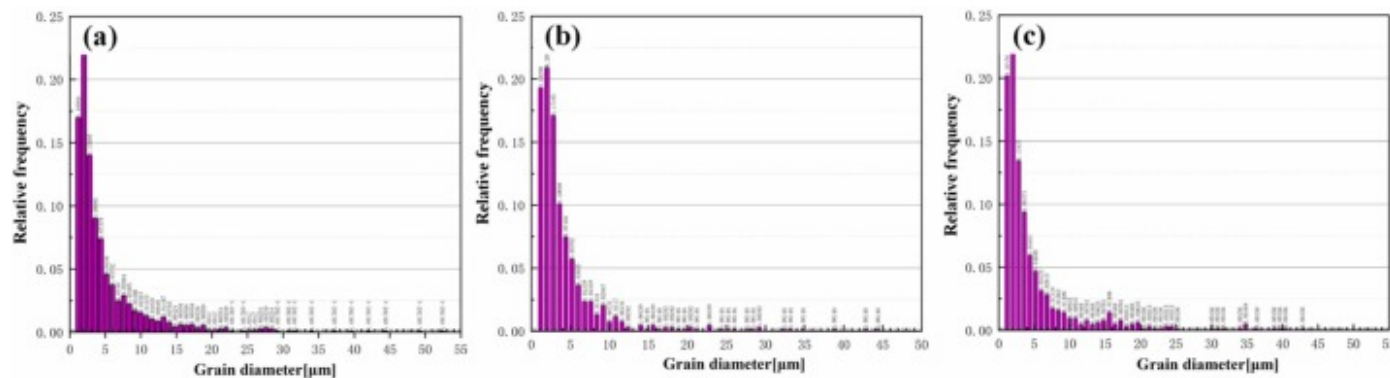
Fig. 19 shows the relative frequency distribution of grain sizes in the WZ, near the fusion line of low-carbon steel, and near the fusion line of medium-carbon steel. Grains less than 5 μm were defined as small grains, and they in the WZ, near the fusion line of low-carbon steel, and near the fusion line of medium-carbon steel accounted for approximately 74%, 75%, and 72%, respectively. Most of the grains in the three regions are small grains. According to the metal strengthening theory [25], the relationship between the grain diameter d and the fine-grain strengthening increment $\Delta\sigma_g$ [26] (Hall-Petch formula), the critical stress increment for crack growth $\Delta\sigma_f$, and the ductile-brittle transition temperature βT_c , are given as follows:

$$\Delta\sigma_g = K_g d^{-1/2} \quad (1)$$

$$\Delta\sigma_f = \frac{2\mu\gamma_p}{K_y} d^{-1/2} \quad (2)$$

$$\beta T_c = \ln B - \ln C - \ln d^{-1/2} \quad (3)$$

where K_g is the grain refinement strengthening coefficient, γ_p is the specific surface energy, K_y is the Petch slope, and μ , β , B , and C are constants. According to formulas (1)–(3), at a smaller grain size d becomes smaller, higher values of $\Delta\sigma_g$ and $\Delta\sigma_f$ and lower values of βT_c decreased, improving the strength and toughness. Moreover, at smaller grain sizes, the number of grain and subgrain boundaries increases, effectively preventing the dislocation formation and dislocation stacking, thereby showing higher macroscopic strength. At the same time, the number of vacancies and dislocations in the grains decreased, and the elastic interaction between dislocations and/or vacancies and dislocations also decreased, as a result, the dislocations moved more easily, thus showing good toughness. Thus, the generation of fine grains in the welded joint plays an important role in improving the strength and toughness of weld bead.

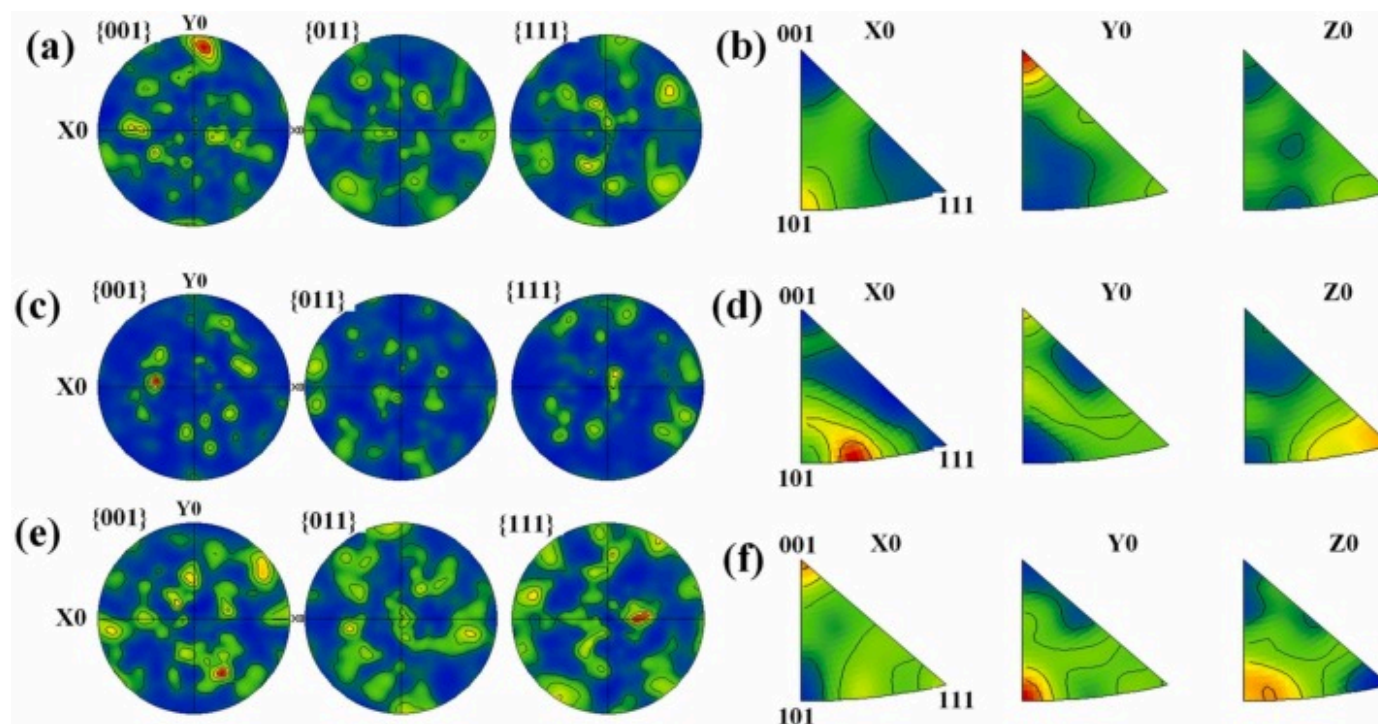


[Download: Download high-res image \(167KB\)](#)

[Download: Download full-size image](#)

Fig. 19. Grain sizes in different regions of the welded joint: (a) WZ; (b) near the fusion line of low-carbon steel; (c) near the fusion line of medium-carbon steel.

Fig. 20 shows the pole figure and the inverse pole figure in the WZ, near the fusion line of low-carbon steel, and in the fusion line area of medium-carbon steel. The polar density intensities of the WZ, near the fusion line of low-carbon steel, and near the fusion line of medium-carbon steel are 2.64, 2.72, and 2.35, respectively. Polar density intensity less than 10 means that there is no texture phenomenon in the growth of the unit cell. Therefore, the polar density intensity value, pole figure, and inverse pole figure of the three regions show that at the three positions, grain growth occurs in any direction; thus, there was no preferred orientation for the growth of the unit cell. Isotropic growth of the unit cell of the welded joint tends to cause a concentrated soft orientation in the welded joint. As a result, the welded joint is prone to shear deformation along the soft orientation direction, showing considerably low tensile strength. The non-oriented growth of the unit cell in the three parts of the welded joint is an important factor for ensuring high strength of the welded joint.



[Download: Download high-res image \(383KB\)](#)

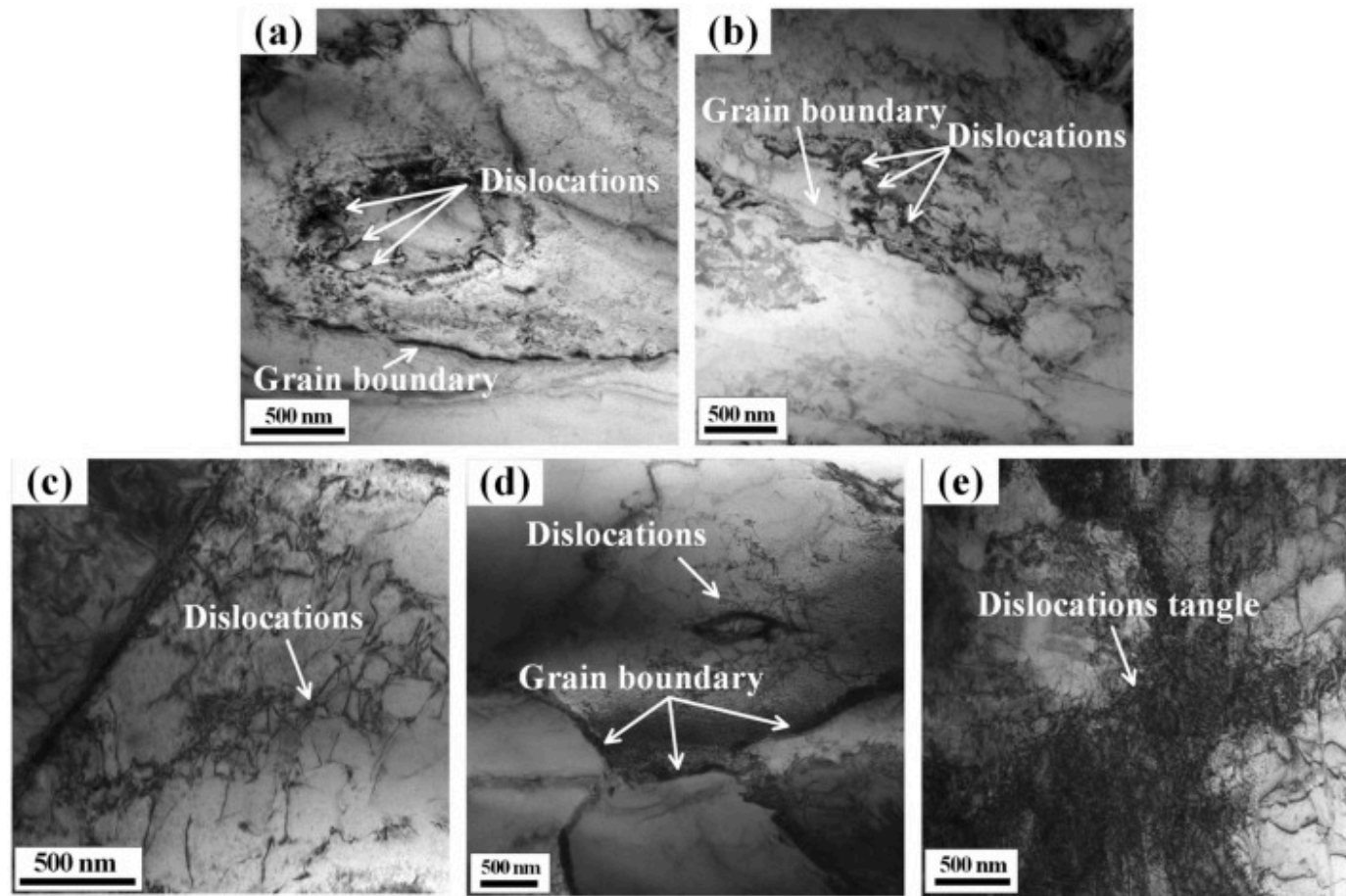
[Download: Download full-size image](#)

Fig. 20. Pole figure and reverse pole figure in different regions of the welded joint: (a, b) WZ; (c, d) near the fusion line of low-carbon steel; (e, f) near the fusion line of medium-carbon steel.

3.4. TEM analysis of welded joint

To further analyze the microstructure of the welded joint at the nanoscale, the welded joint and the base metals were analyzed using HR-TEM. Fig. 21 shows the TEM bright-field images of the base metal area, WZ, low-carbon steel HAZ, and medium-carbon steel HAZ. As shown in Fig. 21 (a) and (b), few dislocation lines are observed in low- as well as medium-carbon steel base metal regions. This is mainly because the grains of the base metal are deformed during the rolling process, resulting in dislocation increments. Comparing the TEM bright-field images of the welded joint and the base metal shows that there are a large number of dislocations and dislocation tangles in the ferrite matrix and at the grain boundaries of the three regions, as shown in Fig. 21 (c)–(e). Many dislocations and dislocation tangles, generated in different regions of the welded joint, are attributed to the

collision of subgrains with varying orientations in the initial stage of crystallization, which further refined the subgrains. The refined subgrains produced dislocation increment in the deformation process, resulting in the accumulation of massive dislocations near the grains and grain boundaries, forming dislocation accumulation. Such a large number of dislocations and dislocation accumulation hinders the occurrence of dislocation movement, which plays a crucial role in improving the strength of the welded joint.

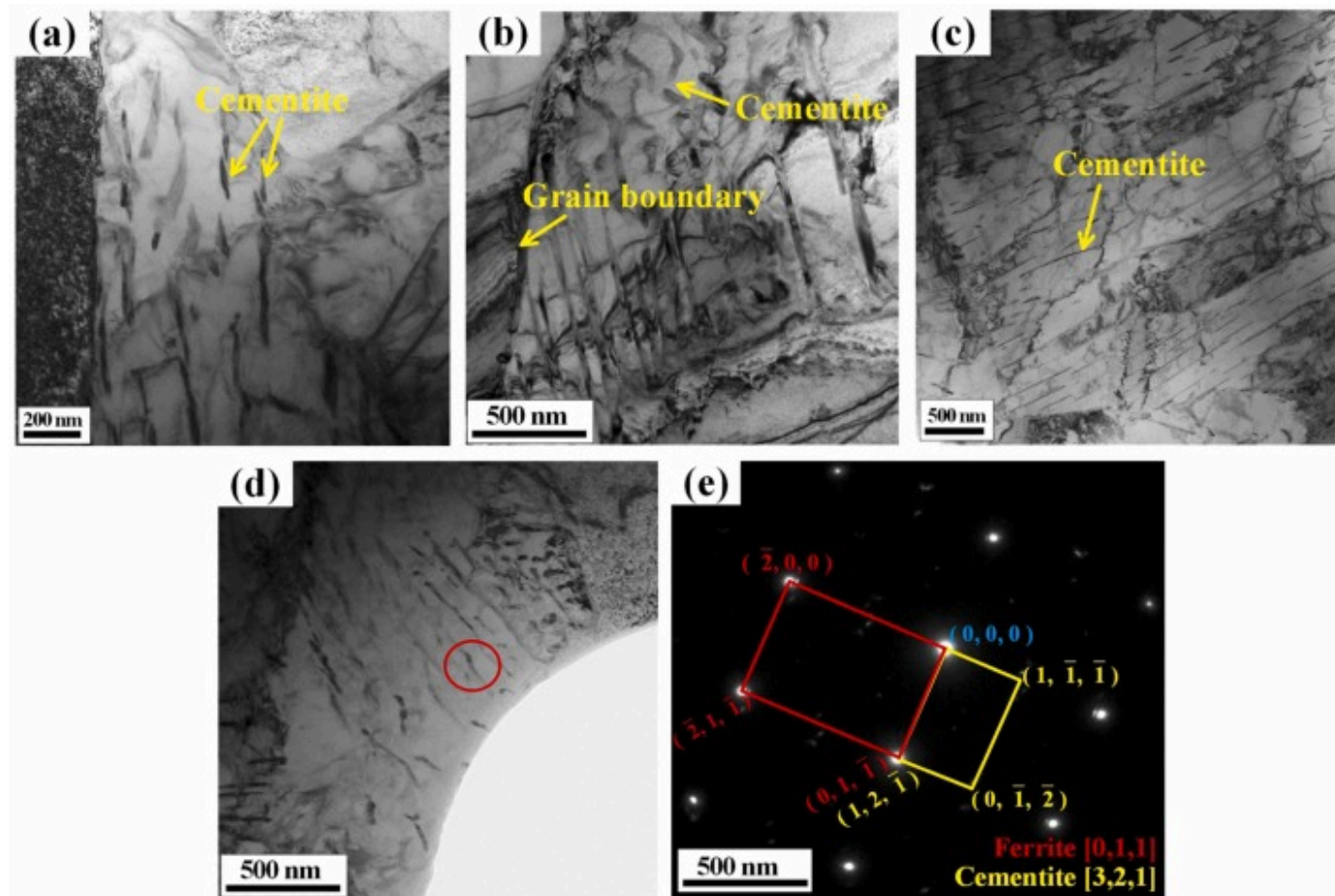


[Download: Download high-res image \(605KB\)](#)

[Download: Download full-size image](#)

Fig. 21. TEM bright-field images of dislocations in the base metal and welded joint: (a) low-carbon steel base metal; (b) medium-carbon steel base metal; (c) WZ; (d) low-carbon steel HAZ; (e) medium-carbon steel HAZ.

As seen in the HR-TEM bright-field images in Fig. 22 (a)–(c), there are many dot and strip carbides in the WZ, low-carbon steel HAZ, and medium-carbon steel HAZ. These dot and strip carbides were identified as cementite by selected area electron diffraction (SAED) analysis (Fig. 22 (e)). Fig. 22 (a) shows that the cementite in the center of the WZ is randomly distributed in the shape of discontinuous dots and strips, so the weld bead is not easily torn along the soft orientation of microstructure during deformation. As a result, the weld bead is strengthened. In the low-carbon steel HAZ, the cementite is distributed in a wave shape (Fig. 22 (b)). The change of cementite morphology may be attributed to the enormous internal stress caused by the widmanstatten structure in the low-carbon steel HAZ. This change in cementite morphology also helps hinder crack formation due to internal stress. In the medium-carbon steel HAZ, the cementite is distributed in layers and sheets in the ferrite matrix (Fig. 22 (c)). The depth of cementite layer is about 10–20 nm, the layer spacing is between 100 and 150 nm, and cementite layers are intermittently distributed. The microstructure of the medium-carbon steel HAZ comprises a large amount of pearlite and a small amount of ferrite. In the pearlite structure, smaller cementite layer spacing and thinner cementite results in higher pearlite hardness and strength.



[Download: Download high-res image \(505KB\)](#)

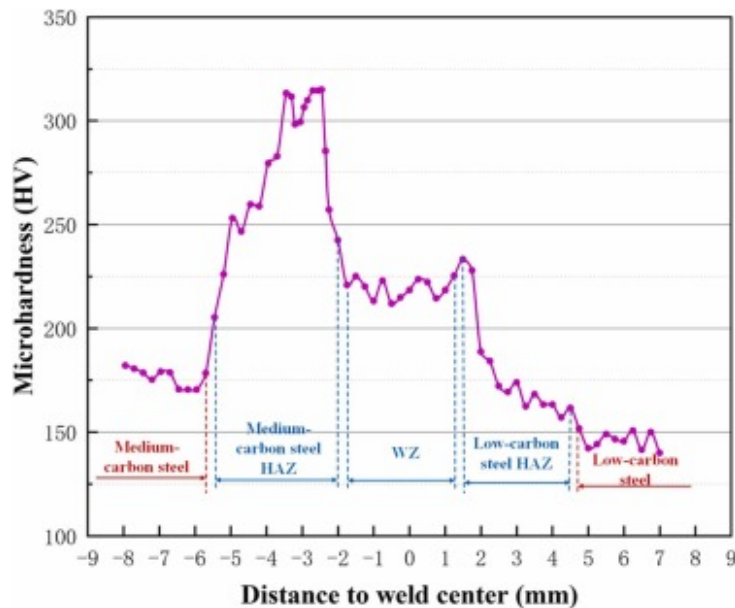
[Download: Download full-size image](#)

Fig. 22. TEM bright-field images and SAED images of cementite in different regions of welded joint: (a) WZ; (b) low-carbon steel HAZ; (c) medium-carbon steel HAZ; (d) bright-field image of cementite; (e) SAED image of cementite in (d).

3.5. Mechanical properties analysis

3.5.1. Hardness analysis

Fig. 23 shows the hardness distribution curve of the welded joint. The hardness distribution of the low- and medium-carbon steel HAZs shows that hardness increased closer to the WZ, and it was the highest near the weld fusion line. The overall hardness distribution curve shows that the hardness of the medium-carbon steel HAZ was higher than that of the WZ, and that of the WZ was higher than that of the low-carbon steel HAZ.



[Download: Download high-res image \(150KB\)](#)

[Download: Download full-size image](#)

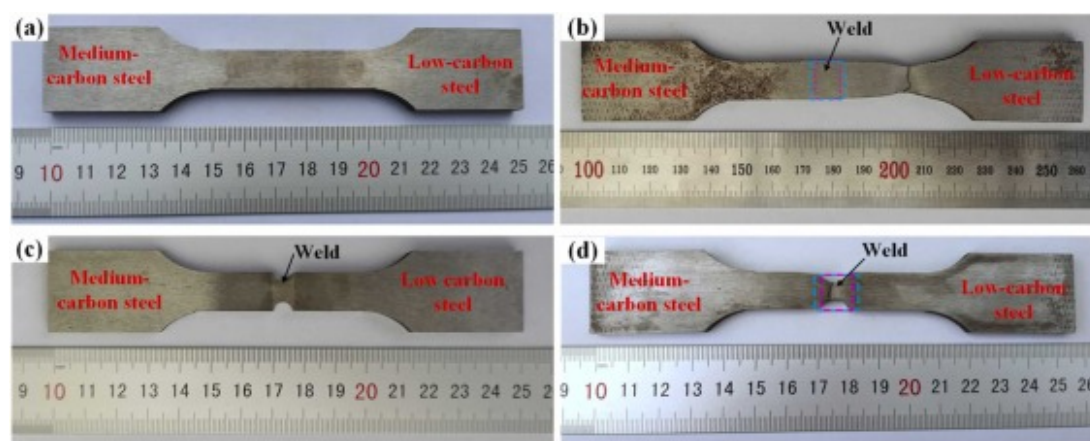
Fig. 23. Hardness curve of welded joint.

Hardness is considerably influenced by the microstructure and grain size. The hardness of the low- and medium-carbon steel HAZs increased closer to the WZ because the amount of high hardness pearlite increased as the two parts of the HAZ reached closer to the WZ. The highest hardness in the low-carbon steel HAZ corresponds to the high hardness widmanstatten structure in the SHZ. The EBSD analysis results near the fusion line of low-carbon steel and the fusion line of medium-carbon steel show that the number of subgrains in the medium-carbon steel HAZ is greater than that in the WZ, and there are more subgrains in the WZ than in the low-carbon steel HAZ. Grain refinement of the material increases the hardness and strength of the material;

therefore, the hardness of the medium-carbon steel HAZ is higher than that of the WZ, and that of the WZ is higher than that of the low-carbon steel HAZ.

3.5.2. Tensile strength analysis

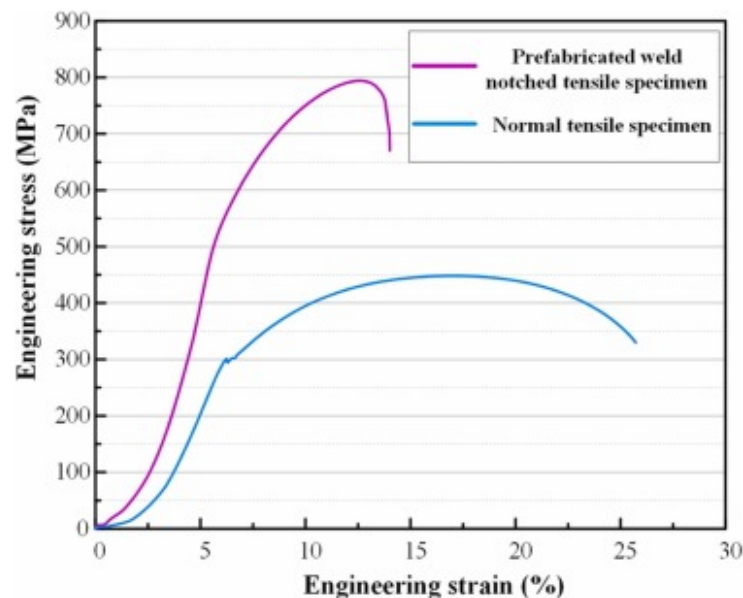
Fig. 24 (a) and (b) show the normal tensile specimens and the fracture samples after stretching. The fracture position of the normal tensile specimen shows that the fracture position occurs at the base metal of low-carbon steel, with tensile strength reaching 450 MPa (Fig. 25). This indicates that the tensile strength of the welded joint is higher than that of the low-carbon steel base metal. There are two main reasons for the improvement of welded joint strength. First, as observed in the EBSD analysis of welded joint, the small grains are mainly distributed near the WZ, low-carbon steel fusion line, and medium-carbon steel fusion line. The distribution of small grains near the welded joint can effectively prevent the occurrence of cracks and dislocation formation, which can improve the strength of the welded joint. Second, as seen from the TEM analysis of the WZ, the low-carbon steel HAZ, and the medium-carbon steel HAZ, dislocation entanglement and dislocation accumulation occur in the grain boundaries and inside the grains in all three regions. A large number of dislocation tangles and dislocation accumulation are also crucial factors for improving the strength of the welded joint.



[Download: Download high-res image \(249KB\)](#)

[Download: Download full-size image](#)

Fig. 24. Photos of tensile specimens before and after stretching: (a) normal tensile specimen; (b) normal tensile specimen after fracture; (c) prefabricated weld notched tensile specimen; (d) prefabricated weld notched tensile specimen after fracture.

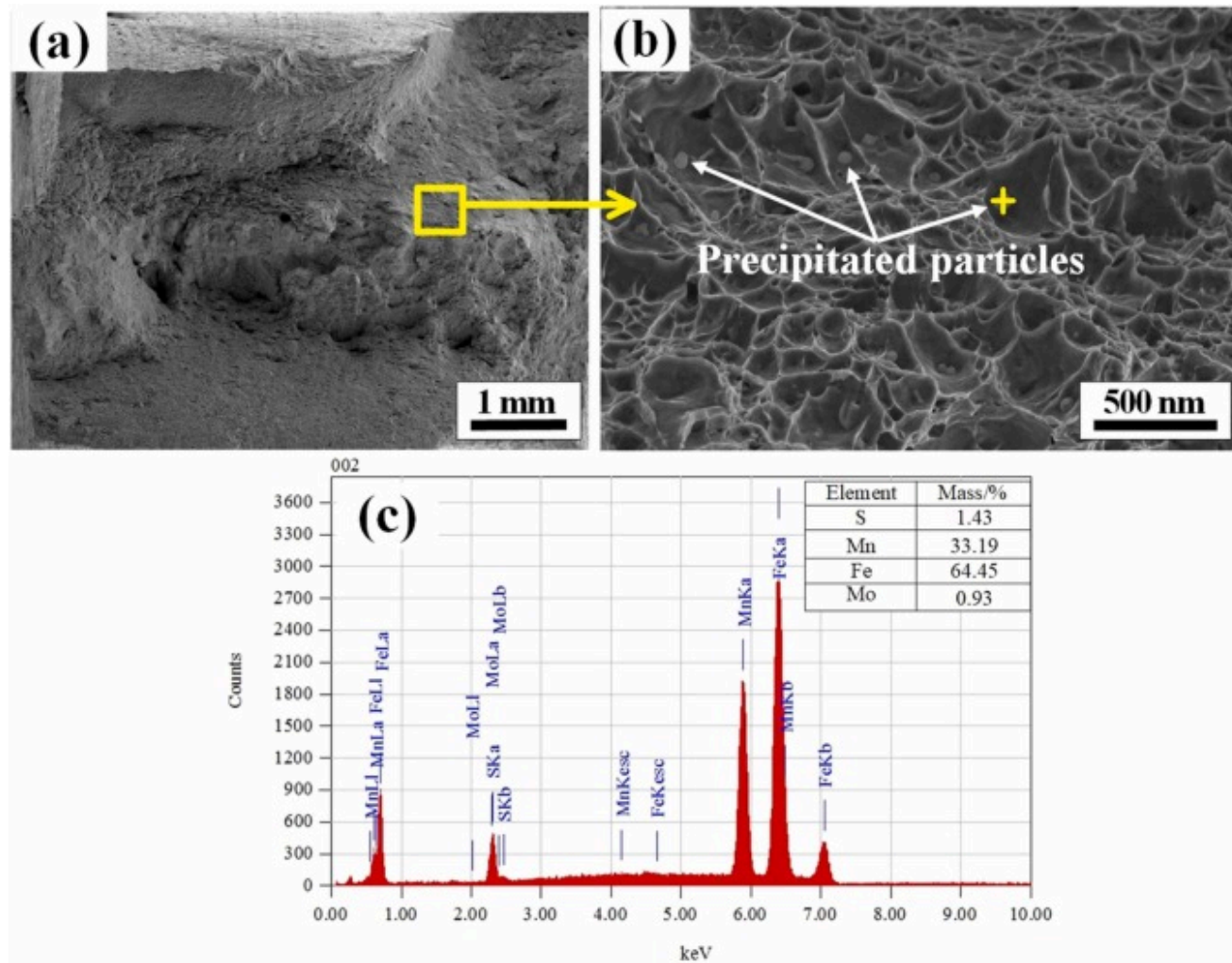


[Download: Download high-res image \(136KB\)](#)

[Download: Download full-size image](#)

Fig. 25. Welded joint tensile curve.

The tensile strength of the weld reached 800 MPa (Fig. 25); the tensile and fracture samples of the weld bead are shown in Fig. 24 (c) and (d). SEM images (Fig. 26 (b)) of the fracture surface show that large dimples of different sizes are distributed; thus, the fracture characteristics of the weld bead are plastic fracture characteristics. In addition, the precipitated particles were distributed at the bottom of the fracture dimples of the tensile specimens. To determine the chemical composition of precipitated particles, EDS analysis was performed on the precipitated particles (Fig. 26 (c)); the precipitated particle mainly comprises S (1.43 wt%), Mn (33.19 wt%), Fe (64.45 wt%), and Mo (0.93 wt%). Therefore, the precipitated particles were mainly MnS particles. The uniform distribution of MnS particles at the bottom of the dimples also contributes to strength improvement of the weld.



[Download: Download high-res image \(526KB\)](#)

[Download: Download full-size image](#)

Fig. 26. SEM image of fractured weld: (a) Macroscopic image of fracture; (b) high-magnification microscopic image of fracture; (c) EDS analysis of precipitated particles.

4. Conclusions

In this study, the effects of laser-arc hybrid welding on the microstructure and properties of low-/medium-carbon steel welded joint are investigated. The findings of the study are as follows:

- (1) By exploring the influence of arc and laser heat inputs on the weld surface and cross-sectional morphology, the optimal combination of welding process parameters was determined. The optimal laser-arc hybrid welding parameters are welding current, arc voltage, laser power, and welding speed of 260 A, 24.8 V, 3300 W, and 0.8 m/min, respectively.
- (2) The microstructure of the WZ comprises ferrite, troostite, upper bainite, and lower bainite. The enrichment of Mn in the WZ promotes the formation of bainite. The microstructure of the low-carbon steel HAZ comprises pearlite, ferrite, and ferrite widmanstatten. The microstructure of the medium-carbon steel HAZ comprises a large amount of pearlite and a small amount of ferrite. In addition, due to the occurrence of the pseudo-eutectic phenomenon in the HAZ in both steels, when the HAZ on both sides is closer to the WZ, higher pearlite content is obtained.
- (3) Small grains (less than 5 μm) are distributed in the WZ, near the fusion line of low-carbon steel, and near the fusion line of medium-carbon steel, and they show no obvious growth orientation. The differences between the microstructure and internal stress in the HAZ and the WZ on both sides change the nanoscopic morphology of cementite and its distribution characteristics. There are a large number of dislocations and dislocation accumulation at grain boundaries in the WZ, low-carbon steel HAZ, and medium-carbon steel HAZ.
- (4) The highest hardness of welded joints occurred in the SHZ of medium-carbon steel; the increase in hardness is attributed to a large amount of pearlite and grain refinement in the SHZ. Tensile tests on the welded joint showed that the strength of the welded joint was higher than that of low-carbon steel base metal. The strength of welded joints improved due to grain refinement strengthening of welded joints and the generation of a large number of dislocation accumulations. Tensile tests on the prefabricated weld notch showed that the weld bead fracture was plastic fracture with a large number of massive dimples.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Jilin Province Science and Technology Development Program project (No. [20200401135GX](#)). The authors would like to thank Changchun Yidong Automobile Parts Manufacturing Co., Ltd. for providing research materials and welding technical support. The authors would also like to thank all the reviewers who gave good suggestions for revisions, as well as MJEditor (www.mjeditor.com) for providing English editing services during the preparation of this manuscript.

[Recommended articles](#)

Data availability

No data was used for the research described in the article.

References

- [1] N. Ghosh, P.K. Pal, G. Nandi
GMAW dissimilar welding of AISI 409 ferritic stainless steel to AISI 316L austenitic stainless steel by using AISI 308 filler wire
Eng. Sci. Technol., Int. J., 20 (2017), pp. 1334-1341
 [View PDF](#) [View article](#) [View in Scopus](#) [Google Scholar](#)
- [2] Y. Javadi, M.C. Smith, K.A. Venkata, N. Naveed, A.N. Forsey, J.A. Francis, R.A. Ainsworth, C.E. Truman, D.J. Smith, F. Hosseinzadeh
Residual stress measurement round robin on an electron beam welded joint between austenitic stainless steel 316L(N) and ferritic steel P91
Int. J. Press. Vessels Pip., 154 (2017), pp. 41-57
 [View PDF](#) [View article](#) [View in Scopus](#) [Google Scholar](#)
- [3] S.K. Albert, C.R. Das, S. Sam, P. Mastanaiah, M. Patel, A.K. Bhaduri, T. Jayakumar, C. Murthy, R. Kumar
Mechanical properties of similar and dissimilar weldments of RAFMS and AISI 316L (N) SS prepared by electron beam welding process
Fusion Eng. Des., 89 (2014), pp. 1605-1610

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [4] X.N. Wang, S.H. Zhang, J. Zhou, M. Zhang, C.J. Chen, R. Misra, Effect of heat input on microstructure and properties of hybrid fiber laser-arc weld joints of the 800MPa hot-rolled Nb-Ti-Mo microalloyed steels, *Optics & Lasers in Engineering*. 91, 2017: 86–96.

[Google Scholar ↗](#)

- [5] F.-x Yin, X.-c Li, C.-x Chen, L. Zhao, Y. Peng, Z.-l Tian
Microstructure and mechanical properties of weld metal in laser and gas metal arc hybrid welding of 440-MPa-grade high-strength steel

J. Iron Steel Res. Int., 28 (2020), pp. 853–861

[Crossref ↗](#) [Google Scholar ↗](#)

- [6] Y. Chen, J. Feng, L. Li, S. Chang, G. Ma
Microstructure and mechanical properties of a thick-section high-strength steel welded joint by novel double-sided hybrid fibre laser-arc welding

Mater. Sci. Eng.: A, 582 (2013), pp. 284–293

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [7] D.J. Smith, G. Zheng, P.R. Hurrell, C.M. Gill, B. Pellereau, K. Ayres, D. Goudar, E. Kingston
Measured and predicted residual stresses in thick section electron beam welded steels

Int. J. Press. Vessels Pip., 120–121 (2014), pp. 66–79

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [8] M. Zhang, Z. Zhang, K. Tang, C. Mao, Y. Hu, G. Chen
Analysis of mechanisms of underfill in full penetration laser welding of thick stainless steel with a 10 kW fiber laser

Opt. Laser Technol., 98 (2018), pp. 97–105

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [9] M. Zhang, G. Chen, Y. Zhou, S. Liao

Optimization of deep penetration laser welding of thick stainless steel with a 10kW fiber laser

Mater. Des., 53 (2014), pp. 568-576



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [10] G. Dak, S. Sirohi, C. Pandey
Study on microstructure and mechanical behavior relationship for laser-welded dissimilar joint of P92 martensitic and 304L austenitic steel

Int. J. Press. Vessels Pip., 196 (2022), Article 104629



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [11] L. Chen, C. Wang, X. Zhang, G. Mi
Effect of parameters on microstructure and mechanical property of dissimilar joints between 316L stainless steel and GH909 alloy by laser welding

J. Manuf. Process., 65 (2021), pp. 60-69



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [12] S. Gao, Y. Li, L. Yang, W. Qiu
Microstructure and mechanical properties of laser-welded dissimilar DP780 and DP980 high-strength steel joints

Mater. Sci. Eng.: A, 720 (2018), pp. 117-129



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [13] Z. Zhang, X. Wang, Q. Sun, B. Yang, L. Xiong, H. Liu
Study on microstructure and properties of laser dissimilar welded joints of ultra-high strength PHS1500/PHS2000 steel

Opt. Laser Technol., 150 (2022), Article 107933



[View PDF](#) [View article](#) [Google Scholar ↗](#)

- [14] B. Acherjee
Hybrid laser arc welding: State-of-art review

Opt. Laser Technol., 99 (2018), pp. 60-71

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [15] G. Ming, X. Zeng, J. Yan, Q. Hu
Microstructure characteristics of laser–MIG hybrid welded mild steel
Appl. Surf. Sci., 254 (2008), pp. 5715-5721
[Google Scholar ↗](#)
- [16] J. Ni, Z. Li, J. Huang, Y. Wu
Strengthening behavior analysis of weld metal of laser hybrid welding for microalloyed steel
Mater. Des., 31 (2010), pp. 4876-4880
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [17] L. Zhiyong, T.S. Srivatsan, L.I. Yan, Z. Wenzhao
Coupling of Laser with Plasma Arc to Facilitate Hybrid Welding of Metallic Materials: A Review
J. Mater. Eng. Perform., 22 (2012), pp. 384-395
[Google Scholar ↗](#)
- [18] X. Zhang, G. Mi, C. Wang
Microstructure and performance of hybrid laser-arc welded high-strength low alloy steel and austenitic stainless steel dissimilar joint - ScienceDirect
Opt. Laser Technol., 122 (2020)
105878-105878
[Google Scholar ↗](#)
- [19] X. Zhou, H. Zhao, F. Liu, B. Yang, B. Chen, C. Tan
Influence of energy ratio on microstructure and mechanical properties in the transition zone of hybrid laser-MIG welded AH36/316L dissimilar joints
J. Mater. Res. Technol., 15 (2021), pp. 4487-4501
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [20] P. Russo Spena, A. Angelastro, G. Casalino

Hybrid laser arc welding of dissimilar TWIP and DP high strength steel weld

J. Manuf. Process., 39 (2019), pp. 233-240



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[21] X. Cao, P. Wanjara, J. Huang, C. Munro, A. Nolting

Hybrid fiber laser – Arc welding of thick section high strength low alloy steel

Mater. Des., 32 (2011), pp. 3399-3413



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[22] S. Wei, S. Lu

Effects of multiple normalizing processes on the microstructure and mechanical properties of low carbon steel weld metal with and without Nb

Mater. Des., 35 (2012), pp. 43-54



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[23] S.K. Liu, J. Zhang

The influence of the Si and Mn concentrations on the kinetics of the bainite transformation in Fe-C-Si-Mn alloys

Metall. Trans. A, 21 (1990), pp. 1517-1525

[View in Scopus ↗](#) [Google Scholar ↗](#)

[24] M. Thompson, M. Ferry, P.A. Manohar

Simulation of Hot-band Microstructure of C-Mn Steels during High Speed Cooling

Isij Int., 41 (2001), pp. 891-899

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[25] H. Huang, P. Zhang, H. Yan, Z. Liu, Z. Yu, D. Wu, H. Shi, Y. Tian

Research on weld formation mechanism of laser-MIG arc hybrid welding with butt gap

Opt. Laser Technol., 133 (2021), Article 106530



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[26] F.B. Pickering, Physical metallurgy and the design of steels, Applied Science Publishers, 1978.

[Google Scholar ↗](#)

Cited by (16)

Induction assisted autogenous plasma arc welding of HSLA steel

2024, Journal of Materials Processing Technology

Citation Excerpt :

...However, the ductile fracture could not able to continue further, and thereafter, there is a propagation of crack from ductile to brittle radial zone. It could be inferred that the larger crack propagation zone contributes a better plastic deformation at the tip which indicates a better impact toughness of the sample [48]. Fig. 13f,h show the fracture surfaces of the radial zone of FZ and CGHAZ....

[Show abstract ✓](#)

Single-pass high-power laser-Arc hybrid welding of thick stainless steel clad plates: Microstructure and mechanical properties

2024, Journal of Materials Research and Technology

Citation Excerpt :

...They found that the corrosion resistance of the weld was comparable to that of the base metal (BM) at a laser-arc distance of 9 mm. It is well known that the heat and mass transfer process during HP-LAHW of thick plates is more intense [17,18]. For example, Zhang et al. [19] studied the microstructure and performance of single-pass HP-LAHW 10 mm EH36 (high-strength low alloy steel) and 316L (austenitic stainless steel) dissimilar joints....

[Show abstract ✓](#)

Study on droplet transfer behavior during medium and low power laser assisted pulsed MAG hybrid welding

2022, Optik

Citation Excerpt :

...Zhang et al. [12] analyzed the impact of CO₂ content in laser arc composite welding shielding gas on welding stability and weld quality. Yang et al. [13] studied the effects of laser arc hybrid welding process parameters on welding quality. The optimum welding parameters were obtained by test, and the weld had good microstructure and mechanical properties under this process parameters....

[Show abstract](#) 

[A comprehensive review of residual stresses in carbon steel welding: formation mechanisms, mitigation strategies, and advanced post-weld heat treatment techniques ↗](#)

2025, International Journal of Advanced Manufacturing Technology

[Study on the Effect of Preheating Temperature on Residual Stress in Laser Arc Composite Welding of Dissimilar Aluminum Alloys ↗](#)

2024, JOM

[Progress and perspectives of joints defects of laser-arc hybrid welding: a review ↗](#)

2024, International Journal of Advanced Manufacturing Technology



[View all citing articles on Scopus ↗](#)

[View Abstract](#)

© 2022 Elsevier GmbH. All rights reserved.

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.